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ROUND-1 MCQs

Section	No. of MCQs
C Programming Fundamentals	40
C++ (OOP, Memory, STL Basics)	35
Pointers & Memory Management	20
Data Structures	30
Algorithms	20
Linux & Unix Commands	20
Software Testing	20
Analytical/Problem Solving	15
Total	200

C Programming Fundamentals MCQs (Q1–Q20)

Q1. Which of the following is a valid C variable declaration?

- A) int 2num;
- B) float salary;
- C) char student-name;
- D) int float;

☒ **Answer: B**

Explanation: Variable names cannot start with a digit, contain -, or be keywords.

Q2. Which header file is required for printf()?

- A) stdlib.h
- B) string.h
- C) stdio.h
- D) math.h

☒ **Answer: C**

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Explanation: printf() is declared in stdio.h.

Q3. Which function is the entry point of every C program?

- A) start()
- B) execute()
- C) main()
- D) run()

☒ **Answer: C**

Explanation: Program execution always begins from main().

Q4. What is the size of char in C?

- A) 2 Bytes
- B) 4 Bytes
- C) 1 Byte
- D) Compiler dependent only

☒ **Answer: C**

Explanation: char is defined to occupy 1 byte in C.

Q5. Which of the following is NOT a valid C keyword?

- A) return
- B) switch
- C) repeat
- D) break

☒ **Answer: C**

Explanation: repeat is not a C keyword.

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Q6. Which data type stores decimal values with single precision?

- A) double
- B) char
- C) float
- D) long

☒ **Answer: C**

Q7. Which symbol is used to end a statement in C?

- A) :
- B) ;
- C) .
- D) ,

☒ **Answer: B**

Q8. What is the output?

```
int x = 10;  
printf("%d", x++);
```

- A) 10
- B) 11
- C) Compilation Error
- D) Garbage Value

☒ **Answer: A**

Explanation: Post-increment prints first, then increments.

Q9. What is the output?

```
int x = 10;  
printf("%d", ++x);
```

- A) 10

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- B) 11
- C) 12
- D) Error

☒ **Answer: B**

Explanation: Pre-increment increments before printing.

Q10. Which operator returns the address of a variable?

- A) *
- B) &
- C) %
- D) #

☒ **Answer: B**

Q11. Which operator is used for dereferencing a pointer?

- A) &
- B) *
- C) %
- D) ->

☒ **Answer: B**

Q12. Which of the following is a correct constant declaration?

- A)
`const int MAX = 100;`
- B)
`int const = 100;`
- C)

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constant int MAX=100;

D)

const MAX=100;

☒ **Answer: A**

Q13. Which loop executes at least once?

A) for

B) while

C) do-while

D) None

☒ **Answer: C**

Explanation: The condition is checked after the first execution.

Q14. Which operator has the highest precedence?

A) +

B) *

C) ()

D) =

☒ **Answer: C**

Q15. Which storage class retains the value of a local variable between function calls?

A) auto

B) register

C) static

D) extern

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✓ **Answer: C**

Q16. What is the output?

```
printf("%d", 5/2);
```

- A) 2.5
- B) 2
- C) 3
- D) Error

✓ **Answer: B**

Explanation: Integer division truncates the decimal part.

Q17. Which function is used to read formatted input from the keyboard?

- A) printf()
- B) scanf()
- C) gets()
- D) puts()

✓ **Answer: B**

Q18. Which of the following is a logical operator?

- A) &&
- B) &
- C) %
- D) <<

✓ **Answer: A**

Q19. What is the default value of an uninitialized local variable?

- A) 0

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B) NULL

C) Garbage Value

D) 1

✓ **Answer: C**

Explanation: Local (automatic) variables are not automatically initialized.

Q20. Which function is used to calculate the length of a string?

A) strcpy()

B) strlen()

C) strcmp()

D) strcat()

✓ **Answer: B**

Explanation: strlen() returns the number of characters in a string (excluding the null terminator).

Q21. Which of the following is NOT a valid storage class in C?

A) auto

B) static

C) register

D) public

✓ **Answer: D**

Explanation: public is not a storage class in C. Valid storage classes include auto, static, register, and extern.

Q22. Which function is used to compare two strings?

A) strcpy()

B) strcmp()

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C) strcat()

D) strlen()

✓ **Answer: B**

Explanation: strcmp() compares two strings and returns 0 if they are equal.

Q23. What is the output?

```
char ch = 'A';  
printf("%d", ch);
```

A) A

B) 65

C) 97

D) Error

✓ **Answer: B**

Explanation: %d prints the ASCII value of 'A', which is 65.

Q24. Which operator is used to access a structure member?

A) *

B) .

C) ->

D) ::

✓ **Answer: B**

Explanation: The dot (.) operator accesses members of a structure variable.

Q25. Which function copies one string into another?

A) strcmp()

B) strcpy()

C) strcat()

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D) strlen()

✓ **Answer: B**

Explanation: strcpy() copies the source string to the destination string.

Q26. Which escape sequence is used for a new line?

A) \t

B) \n

C) \r

D) \b

✓ **Answer: B**

Q27. Which loop is most suitable when the number of iterations is known in advance?

A) while

B) do-while

C) for

D) goto

✓ **Answer: C**

Q28. Which keyword is used to skip the remaining statements in the current loop iteration?

A) break

B) continue

C) exit

D) return

✓ **Answer: B**

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Q29. Which keyword immediately terminates a loop?

- A) continue
- B) stop
- C) break
- D) skip

☒ **Answer: C**

Q30. What is the output?

```
int a = 5;
```

```
if(a > 3)
    printf("Yes");
else
    printf("No");
```

- A) Yes
- B) No
- C) Error
- D) No Output

☒ **Answer: A**

Q31. Which library function is used to dynamically allocate memory for a single block?

- A) calloc()
- B) realloc()
- C) malloc()
- D) free()

☒ **Answer: C**

Explanation: malloc() allocates a block of memory but does not initialize it.

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Q32. Which function is used to release dynamically allocated memory?

- A) delete()
- B) remove()
- C) free()
- D) release()

☒ **Answer: C**

Q33. Which header file is required for malloc()?

- A) stdio.h
- B) string.h
- C) stdlib.h
- D) math.h

☒ **Answer: C**

Q34. Which keyword is used to define a user-defined data type?

- A) typedef
- B) define
- C) structonly
- D) alias

☒ **Answer: A**

Q35. What is the output?

```
int a = 10;  
printf("%d", a % 3);
```

- A) 0
- B) 1

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C) 2

D) 3

✓ **Answer: B**

Explanation: The remainder when 10 is divided by 3 is 1.

Q36. Which statement is used to return a value from a function?

A) break

B) continue

C) return

D) exit

✓ **Answer: C**

Q37. Which operator is used for logical OR in C?

A) |

B) ||

C) &

D) &&

✓ **Answer: B**

Q38. Which keyword is used to declare a global variable defined in another file?

A) static

B) auto

C) extern

D) register

✓ **Answer: C**

Explanation: `extern` tells the compiler that the variable is defined elsewhere.

Q39. What is the output?

```
int x = 3;
```

```
switch(x)
{
    case 1: printf("One"); break;
    case 2: printf("Two"); break;
    case 3: printf("Three"); break;
    default: printf("Default");
}
```

- A) One
- B) Two
- C) Three
- D) Default

✓ **Answer: C**

Q40. Which of the following statements about C is correct?

- A) C supports multiple inheritance.
- B) C is an object-oriented language.
- C) C is a procedural programming language.
- D) C has built-in garbage collection.

✓ **Answer: C**

Explanation: C is a procedural programming language. Object-oriented features and garbage collection are not part of standard C.

C++ (OOP, Memory, STL Basics) MCQs (Q1–Q35)

Q1. C++ is an extension of which programming language?

- A) Java

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B) Python

C) C

D) Pascal

☒ **Answer: C**

Q2. Which feature is NOT supported by C?

A) Functions

B) Pointers

C) Classes

D) Arrays

☒ **Answer: C**

Q3. Which of the following is the correct way to define a class?

A)

```
class Student {};
```

B)

```
Student class {};
```

C)

```
define Student {};
```

D)

```
new class Student {};
```

☒ **Answer: A**

Q4. What is an object?

A) Blueprint

B) Instance of a class

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C) Function

D) Variable

✓ **Answer: B**

Q5. Which OOP concept binds data and functions together?

A) Inheritance

B) Encapsulation

C) Polymorphism

D) Abstraction

✓ **Answer: B**

Q6. Which keyword is used to access members of an object?

A) ::

B) .

C) ->

D) *

✓ **Answer: B**

Q7. Which member function is automatically called when an object is created?

A) Destructor

B) Constructor

C) Friend Function

D) Static Function

✓ **Answer: B**

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Q8. Which member function is automatically called when an object is destroyed?

- A) Constructor
- B) Friend Function
- C) Destructor
- D) Static Function

 **Answer: C**

Q9. What is the name of a constructor having no parameters?

- A) Copy Constructor
- B) Parameterized Constructor
- C) Default Constructor
- D) Virtual Constructor

 **Answer: C**

Q10. Which constructor initializes an object using another object?

- A) Copy Constructor
- B) Default Constructor
- C) Friend Constructor
- D) Static Constructor

 **Answer: A**

Q11. Which keyword is used to allocate memory dynamically in C++?

- A) malloc
- B) alloc
- C) new

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D) create

✓ **Answer: C**

Q12. Which keyword is used to release memory allocated using new?

A) free

B) release

C) delete

D) remove

✓ **Answer: C**

Q13. Which operator accesses members through an object pointer?

A) .

B) ::

C) ->

D) *

✓ **Answer: C**

Q14. Which OOP feature allows one class to acquire properties of another?

A) Encapsulation

B) Inheritance

C) Abstraction

D) Overloading

✓ **Answer: B**

Q15. Which inheritance mode keeps public members public?

A) Public

B) Private

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C) Protected

D) Static

☒ **Answer: A**

Q16. Function overloading is an example of?

A) Runtime Polymorphism

B) Compile-time Polymorphism

C) Inheritance

D) Encapsulation

☒ **Answer: B**

Q17. Which keyword enables runtime polymorphism?

A) friend

B) static

C) virtual

D) inline

☒ **Answer: C**

Q18. Which function can be overridden in a derived class?

A) Constructor

B) Virtual Function

C) Friend Function

D) Static Function

☒ **Answer: B**

Q19. Which keyword prevents inheritance?

A) final

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B) const

C) virtual

D) static

☒ **Answer: A**

Q20. Which keyword refers to the current object?

A) self

B) object

C) this

D) current

☒ **Answer: C**

Q21. Which access specifier makes members accessible only inside the class?

A) public

B) protected

C) private

D) global

☒ **Answer: C**

Q22. Which access specifier allows derived classes to access members?

A) private

B) protected

C) public only

D) static

☒ **Answer: B**

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Q23. Which function can access private members without being a class member?

- A) Static Function
- B) Virtual Function
- C) Friend Function
- D) Inline Function

 **Answer: C**

Q24. Which keyword is used to define a constant member function?

- A) static
- B) virtual
- C) const
- D) final

 **Answer: C**

Q25. Which STL container stores elements in contiguous memory?

- A) list
- B) vector
- C) map
- D) set

 **Answer: B**

Q26. Which STL container follows the LIFO principle?

- A) queue
- B) stack
- C) deque

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D) vector

✓ **Answer: B**

Q27. Which STL container follows the FIFO principle?

A) stack

B) vector

C) queue

D) map

✓ **Answer: C**

Q28. Which STL container stores unique sorted keys?

A) vector

B) set

C) list

D) queue

✓ **Answer: B**

Q29. Which STL container stores key-value pairs?

A) vector

B) stack

C) map

D) queue

✓ **Answer: C**

Q30. Which header file is required for vector?

A) <array>

B) <vector>

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C) <list>

D) <map>

☒ **Answer: B**

Q31. Which STL algorithm sorts elements?

A) reverse()

B) sort()

C) unique()

D) swap()

☒ **Answer: B**

Q32. Which header file contains sort()?

A) <vector>

B) <algorithm>

C) <string>

D) <utility>

☒ **Answer: B**

Q33. What is the purpose of a destructor?

A) Allocate memory

B) Initialize data

C) Release resources before object destruction

D) Copy an object

☒ **Answer: C**

Q34. Which statement about virtual functions is correct?

A) They enable compile-time polymorphism.

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- B) They enable runtime polymorphism.
- C) They cannot be inherited.
- D) They are constructors.

☒ **Answer: B**

Q35. Which feature hides implementation details and exposes only essential functionality?

- A) Inheritance
- B) Abstraction
- C) Encapsulation
- D) Overloading

☒ **Answer: B**

Q41. What is a pointer in C?

- A) A keyword
- B) A variable that stores the address of another variable
- C) A function
- D) A constant

☒ **Answer: B**

Explanation: A pointer stores the memory address of another variable.

Q42. Which declaration correctly defines a pointer to an integer?

- A) int ptr;
- B) int *ptr;
- C) pointer int ptr;
- D) *int ptr;

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✓ **Answer: B**

Q43. What does the & operator do?

- A) Returns the value stored in a pointer
- B) Returns the address of a variable
- C) Performs logical AND
- D) Performs multiplication

✓ **Answer: B**

Q44. What does the * operator do when used with a pointer?

- A) Returns the memory address
- B) Dereferences the pointer to access the value
- C) Performs multiplication
- D) Declares a constant

✓ **Answer: B**

Q45. What is the output?

```
int a = 10;  
int *p = &a;  
printf("%d", *p);
```

- A) Address of a
- B) 10
- C) Garbage value
- D) Compilation Error

✓ **Answer: B**

Q46. Which function allocates memory but does NOT initialize it?

- A) calloc()

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B) malloc()

C) realloc()

D) free()

☒ **Answer: B**

Q47. Which function allocates memory and initializes it to zero?

A) malloc()

B) calloc()

C) realloc()

D) memset()

☒ **Answer: B**

Q48. Which function changes the size of an already allocated memory block?

A) malloc()

B) calloc()

C) realloc()

D) free()

☒ **Answer: C**

Q49. Which function is used to release dynamically allocated memory?

A) delete

B) dispose

C) free()

D) remove()

☒ **Answer: C**

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Q50. What is a NULL pointer?

- A) Pointer storing value 1
- B) Pointer pointing to address 0 / no valid object
- C) Uninitialized pointer
- D) Pointer to a character

☒ **Answer: B**

Q51. What is a wild pointer?

- A) Pointer initialized to NULL
- B) Pointer that has not been initialized before use
- C) Pointer returned by malloc()
- D) Pointer to a function

☒ **Answer: B**

Q52. What is a dangling pointer?

- A) Pointer pointing to valid memory
- B) Pointer referring to memory that has been freed or is no longer valid
- C) Pointer initialized to NULL
- D) Pointer to an array

☒ **Answer: B**

Q53. What is the safest practice immediately after calling free(ptr)?

- A) Increment the pointer
- B) Set the pointer to NULL
- C) Print the pointer
- D) Dereference the pointer

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✓ **Answer: B**

Explanation: Setting it to NULL helps avoid accidental use of a dangling pointer.

Q54. What is a memory leak?

- A) Memory allocated but never released
- B) Accessing an array element
- C) Using a global variable
- D) Printing an address

✓ **Answer: A**

Q55. Which expression accesses the second element using pointer arithmetic?

```
int arr[5];  
int *p = arr;
```

- A) *p
- B) *(p + 1)
- C) *(p - 1)
- D) &p

✓ **Answer: B**

Q56. What is the output?

```
int x = 5;  
int *p = &x;  
*p = 20;  
printf("%d", x);
```

- A) 5
- B) 20
- C) Garbage value
- D) Error

✓ **Answer: B**

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Q57. Which pointer can point to any data type?

- A) char *
- B) void *
- C) int *
- D) float *

☒ **Answer: B**

Q58. What is the output?

```
int arr[] = {10,20,30};  
printf("%d", *(arr + 2));
```

- A) 10
- B) 20
- C) 30
- D) Error

☒ **Answer: C**

Q59. Which statement about arrays and pointers is correct?

- A) Arrays and pointers are identical.
- B) An array name usually decays to a pointer to its first element in expressions.
- C) A pointer cannot point to an array element.
- D) Arrays can always be assigned like pointers.

☒ **Answer: B**

Explanation: Arrays and pointers are closely related but are not the same thing.

Q60. Which of the following can cause undefined behavior?

- A) Dereferencing a NULL pointer
- B) Using printf()

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- C) Declaring an integer
- D) Returning 0 from main()

☒ **Answer: A**

LG Soft India – Data Structures MCQs (Q1–Q30)

Q1. What is a Data Structure?

- A) A programming language
- B) A method of organizing and storing data efficiently
- C) A compiler
- D) An operating system

☒ **Answer: B**

Q2. Which of the following is a Linear Data Structure?

- A) Tree
- B) Graph
- C) Array
- D) Heap

☒ **Answer: C**

Q3. Which of the following is a Non-Linear Data Structure?

- A) Stack
- B) Queue
- C) Linked List
- D) Tree

☒ **Answer: D**

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Q4. Which data structure stores elements in contiguous memory locations?

- A) Linked List
- B) Array
- C) Queue
- D) Stack

☒ **Answer: B**

Q5. Which data structure allows insertion and deletion at both ends?

- A) Queue
- B) Stack
- C) Deque
- D) Array

☒ **Answer: C**

Q6. Which data structure follows the LIFO principle?

- A) Queue
- B) Stack
- C) Linked List
- D) Tree

☒ **Answer: B**

Q7. Which data structure follows the FIFO principle?

- A) Queue
- B) Stack
- C) Tree
- D) Graph

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✓ Answer: A

Q8. Which operation inserts an element into a stack?

- A) Insert
- B) Push
- C) Enqueue
- D) Add

✓ Answer: B

Q9. Which operation removes an element from a stack?

- A) Delete
- B) Remove
- C) Pop
- D) Dequeue

✓ Answer: C

Q10. Which operation inserts an element into a queue?

- A) Push
- B) Insert
- C) Enqueue
- D) Pop

✓ Answer: C

Q11. Which operation removes an element from a queue?

- A) Delete
- B) Pop
- C) Remove

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D) Dequeue

✓ **Answer: D**

Q12. Which data structure uses pointers to connect nodes?

A) Array

B) Linked List

C) Stack (Array Implementation)

D) Matrix

✓ **Answer: B**

Q13. Which linked list allows traversal in both directions?

A) Singly Linked List

B) Circular Linked List

C) Doubly Linked List

D) Static List

✓ **Answer: C**

Q14. Which linked list has the last node connected to the first node?

A) Doubly Linked List

B) Circular Linked List

C) Singly Linked List

D) Binary Tree

✓ **Answer: B**

Q15. Which searching algorithm requires sorted data?

A) Linear Search

B) Binary Search

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C) Hash Search

D) Sequential Search

✓ **Answer: B**

Q16. What is the worst-case time complexity of Linear Search?

A) $O(1)$

B) $O(\log n)$

C) $O(n)$

D) $O(n^2)$

✓ **Answer: C**

Q17. What is the worst-case time complexity of Binary Search?

A) $O(n)$

B) $O(\log n)$

C) $O(n \log n)$

D) $O(n^2)$

✓ **Answer: B**

Q18. Which traversal visits Root → Left → Right?

A) Inorder

B) Preorder

C) Postorder

D) Level Order

✓ **Answer: B**

Q19. Which traversal visits Left → Root → Right?

A) Preorder

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- B) Postorder
- C) Inorder
- D) Level Order

☒ **Answer: C**

Q20. Which traversal visits Left → Right → Root?

- A) Inorder
- B) Preorder
- C) Postorder
- D) Breadth First

☒ **Answer: C**

Q21. Which data structure is primarily used to implement recursion?

- A) Queue
- B) Stack
- C) Heap
- D) Graph

☒ **Answer: B**

Q22. Which data structure is commonly used for Breadth-First Search (BFS)?

- A) Stack
- B) Queue
- C) Array
- D) Linked List

☒ **Answer: B**

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Q23. Which data structure is commonly used for Depth-First Search (DFS)?

- A) Queue
- B) Stack
- C) Heap
- D) Array

☒ **Answer: B**

Q24. Which tree is used for fast searching, insertion, and deletion of ordered data?

- A) Binary Search Tree (BST)
- B) General Tree
- C) Expression Tree
- D) Trie

☒ **Answer: A**

Q25. Which data structure stores data in key-value pairs for fast lookup?

- A) Stack
- B) Queue
- C) Hash Table
- D) Array

☒ **Answer: C**

Q26. Which collision resolution technique uses another location in the table when two keys map to the same index?

- A) Overflow
- B) Hashing

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C) Open Addressing

D) Sorting

✓ **Answer: C**

Q27. Which data structure is best suited for implementing a browser's Back button?

A) Queue

B) Stack

C) Tree

D) Graph

✓ **Answer: B**

Q28. Which data structure is best suited for CPU scheduling using First Come First Serve (FCFS)?

A) Stack

B) Queue

C) Linked List

D) Tree

✓ **Answer: B**

Q29. Which notation represents the upper bound of an algorithm's running time?

A) Big Ω (Omega)

B) Big Θ (Theta)

C) Big O

D) Small o

✓ **Answer: C**

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Q30. Which of the following data structures is generally the most memory-efficient for storing a sparse graph?

- A) Adjacency Matrix
- B) Adjacency List
- C) 2D Array
- D) Complete Binary Tree

 **Answer: B**

Explanation: An adjacency list stores only existing edges, making it much more memory-efficient than an adjacency matrix for sparse graphs.

LG Soft India – Algorithms MCQs (Q1–Q20)

Q1. What is an algorithm?

- A) A programming language
- B) A finite sequence of well-defined steps to solve a problem
- C) A compiler
- D) A data structure

 **Answer: B**

Explanation: An algorithm is a step-by-step procedure for solving a problem.

Q2. Which characteristic is NOT a property of a good algorithm?

- A) Finiteness
- B) Definiteness
- C) Ambiguity
- D) Effectiveness

 **Answer: C**

Explanation: Every step in an algorithm must be clear and unambiguous.

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Q3. Which notation represents the worst-case time complexity?

- A) Big Ω
- B) Big Θ
- C) Big O
- D) Little o

☒ **Answer: C**

Q4. What is the time complexity of Linear Search in the worst case?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

☒ **Answer: C**

Q5. Binary Search works only on

- A) Random Data
- B) Sorted Data
- C) Linked List Only
- D) Graph

☒ **Answer: B**

Q6. Worst-case time complexity of Binary Search is

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(1)$

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✓ Answer: B

Q7. Which sorting algorithm repeatedly swaps adjacent elements?

- A) Merge Sort
- B) Bubble Sort
- C) Quick Sort
- D) Heap Sort

✓ Answer: B

Q8. Best-case time complexity of Bubble Sort (optimized version) is

- A) $O(n^2)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n \log n)$

✓ Answer: B

Q9. Which sorting algorithm follows the Divide and Conquer approach?

- A) Bubble Sort
- B) Selection Sort
- C) Merge Sort
- D) Insertion Sort

✓ Answer: C

Q10. Average-case time complexity of Quick Sort is

- A) $O(n^2)$
- B) $O(\log n)$
- C) $O(n \log n)$

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D) $O(n)$

✓ **Answer: C**

Q11. Worst-case time complexity of Quick Sort is

A) $O(n^2)$

B) $O(\log n)$

C) $O(n)$

D) $O(n \log n)$

✓ **Answer: A**

Explanation: This occurs when the pivot selection is consistently poor (e.g., always the smallest or largest element in an already sorted array).

Q12. Which sorting algorithm always has $O(n \log n)$ time complexity in the worst case?

A) Bubble Sort

B) Quick Sort

C) Merge Sort

D) Insertion Sort

✓ **Answer: C**

Q13. Which sorting algorithm selects the minimum element in every pass?

A) Bubble Sort

B) Selection Sort

C) Merge Sort

D) Heap Sort

✓ **Answer: B**

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Q14. Which sorting algorithm inserts each element into its correct position?

- A) Bubble Sort
- B) Merge Sort
- C) Insertion Sort
- D) Quick Sort

 **Answer: C**

Q15. Which algorithmic technique solves a problem by breaking it into smaller subproblems and combining the results?

- A) Greedy Method
- B) Divide and Conquer
- C) Backtracking
- D) Brute Force

 **Answer: B**

Q16. Which algorithmic technique makes the best local choice at each step?

- A) Divide and Conquer
- B) Dynamic Programming
- C) Greedy Algorithm
- D) Backtracking

 **Answer: C**

Q17. Which algorithmic technique stores solutions to overlapping subproblems to avoid recomputation?

- A) Greedy
- B) Dynamic Programming

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- C) Recursion Only
- D) Divide and Conquer

✓ **Answer: B**

Q18. Which algorithmic technique explores all possible solutions and backtracks when necessary?

- A) Greedy
- B) Dynamic Programming
- C) Backtracking
- D) Binary Search

✓ **Answer: C**

Q19. Which searching algorithm is generally preferred for a small, unsorted array?

- A) Binary Search
- B) Linear Search
- C) Jump Search
- D) Interpolation Search

✓ **Answer: B**

Explanation: Linear Search does not require the data to be sorted and is simple to implement for small datasets.

Q20. Which of the following has the fastest growth rate as the input size increases?

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n \log n)$

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D) $O(n^2)$

✓ **Answer: D**

Explanation: Among these options, $O(n^2)$ grows the fastest as n increases.

Linux & Unix Commands MCQs (Q1–Q20)

Q1. Which command is used to display the current working directory?

A) pwd

B) cd

C) ls

D) dir

✓ **Answer: A**

Explanation: pwd (Print Working Directory) displays the absolute path of the current directory.

Q2. Which command is used to list files and directories?

A) pwd

B) ls

C) cat

D) mkdir

✓ **Answer: B**

Q3. Which command is used to change the current directory?

A) mv

B) cp

C) cd

D) pwd

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✓ **Answer: C**

Q4. Which command creates a new directory?

- A) touch
- B) mkdir
- C) rmdir
- D) mkfile

✓ **Answer: B**

Q5. Which command removes an empty directory?

- A) rm
- B) delete
- C) rmdir
- D) erase

✓ **Answer: C**

Q6. Which command creates an empty file?

- A) touch
- B) new
- C) create
- D) file

✓ **Answer: A**

Q7. Which command displays the contents of a text file?

- A) cat
- B) grep
- C) pwd

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D) ls

✓ **Answer: A**

Q8. Which command copies a file?

A) mv

B) cp

C) copy

D) clone

✓ **Answer: B**

Q9. Which command renames or moves a file?

A) rename

B) cp

C) mv

D) move

✓ **Answer: C**

Explanation: mv is used both to move files and to rename them.

Q10. Which command permanently removes a file?

A) delete

B) erase

C) rm

D) removefile

✓ **Answer: C**

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Q11. Which command searches for a specific pattern inside a file?

- A) find
- B) grep
- C) locate
- D) where

☒ **Answer: B**

Q12. Which command changes file permissions?

- A) chmod
- B) chown
- C) chgrp
- D) perm

☒ **Answer: A**

Q13. Which command changes the owner of a file?

- A) chmod
- B) chown
- C) chgrp
- D) owner

☒ **Answer: B**

Q14. What does the command `ls -l` display?

- A) Hidden files only
- B) Long listing format with permissions, owner, size, and date
- C) Only directories
- D) System files

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☒ **Answer: B**

Q15. Which command displays all running processes?

A) jobs

B) ps

C) kill

D) pwd

☒ **Answer: B**

Q16. Which command is commonly used to terminate a running process using its Process ID (PID)?

A) stop

B) exit

C) kill

D) close

☒ **Answer: C**

Q17. Which command displays real-time information about running processes and system resource usage?

A) ps

B) jobs

C) top

D) who

☒ **Answer: C**

Q18. Which command displays the current logged-in user's username?

A) who

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B) whoami

C) id

D) users

✓ **Answer: B**

Q19. What is the meaning of the permission `rwxr-xr--`?

A) Owner: Read, Write, Execute; Group: Read, Execute; Others: Read

B) Everyone has full permissions

C) Owner has read only

D) Group has write permission

✓ **Answer: A**

Explanation:

- Owner: `rw` = Read, Write, Execute
- Group: `r-x` = Read, Execute
- Others: `r--` = Read only

Q20. Which command is used to find files by name in a directory hierarchy?

A) `grep`

B) `find`

C) `search`

D) `where`

✓ **Answer: B**

Explanation: `find` searches recursively through directories based on criteria such as name, type, or size.

Software Testing MCQs (Q1–Q20)

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Q1. What is Software Testing?

- A) Writing source code
- B) The process of verifying and validating software to identify defects
- C) Installing software
- D) Designing databases

 **Answer: B**

Explanation: Software testing verifies that software meets requirements and helps identify defects before release.

Q2. Which testing is performed without executing the program?

- A) Dynamic Testing
- B) Static Testing
- C) Regression Testing
- D) Integration Testing

 **Answer: B**

Explanation: Static testing includes reviews, inspections, and static code analysis without executing the software.

Q3. Which testing is performed by executing the application?

- A) Static Testing
- B) Dynamic Testing
- C) Code Review
- D) Walkthrough

 **Answer: B**

Q4. What is the main objective of Software Testing?

- A) To prove software is error-free

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- B) To identify defects and ensure software meets requirements
- C) To increase file size
- D) To reduce coding time

☒ **Answer: B**

Q5. Which document defines the expected behavior of software?

- A) Source Code
- B) Software Requirement Specification (SRS)
- C) Log File
- D) Test Report

☒ **Answer: B**

Q6. Which SDLC phase follows the Design phase?

- A) Requirement Analysis
- B) Coding (Implementation)
- C) Maintenance
- D) Testing

☒ **Answer: B**

Q7. What does STLC stand for?

- A) Software Testing Life Cycle
- B) Software Technical Life Cycle
- C) System Test Level Cycle
- D) Software Technology Life Cycle

☒ **Answer: A**

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Q8. Which phase comes first in the STLC?

- A) Test Execution
- B) Test Planning
- C) Test Closure
- D) Defect Reporting

✓ **Answer: B**

Explanation: Test planning defines scope, objectives, resources, schedule, and strategy before execution.

Q9. What is a Test Case?

- A) A programming language
- B) A document containing inputs, execution conditions, and expected results
- C) A bug report
- D) A database table

✓ **Answer: B**

Q10. Which testing verifies individual modules or functions?

- A) System Testing
- B) Unit Testing
- C) Acceptance Testing
- D) Regression Testing

✓ **Answer: B**

Q11. Which testing verifies the interaction between integrated modules?

- A) Unit Testing
- B) Integration Testing

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C) Smoke Testing

D) Alpha Testing

✓ **Answer: B**

Q12. Which testing verifies the complete application as a whole?

A) Unit Testing

B) Integration Testing

C) System Testing

D) Component Testing

✓ **Answer: C**

Q13. Which testing is performed by the customer or end user before final release?

A) Unit Testing

B) Integration Testing

C) User Acceptance Testing (UAT)

D) Regression Testing

✓ **Answer: C**

Q14. What is Regression Testing?

A) Testing only new features

B) Re-testing the software after changes to ensure existing functionality still works

C) Testing hardware

D) Testing documentation

✓ **Answer: B**

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Q15. What is Smoke Testing?

- A) Detailed testing of every feature
- B) Preliminary testing to verify that the critical functionalities work and the build is stable
- C) Performance testing
- D) Security testing

☒ **Answer: B**

Q16. What is Black Box Testing?

- A) Testing based on internal source code
- B) Testing functionality without knowledge of internal implementation
- C) Kernel testing
- D) Database testing

☒ **Answer: B**

Q17. What is White Box Testing?

- A) Testing only the user interface
- B) Testing based on internal code structure and logic
- C) Network testing
- D) Browser testing

☒ **Answer: B**

Q18. What is a Defect (Bug)?

- A) A software feature
- B) A deviation between expected and actual behavior
- C) A hardware component

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D) A compiler warning

✓ **Answer: B**

Q19. Which of the following is a valid bug life cycle stage?

A) New

B) Assigned

C) Fixed

D) All of the Above

✓ **Answer: D**

Explanation: A typical defect life cycle includes stages such as New, Assigned, Open, Fixed, Retest, Verified, Closed, and Reopened (the exact workflow may vary by organization).

Q20. Which testing technique divides input data into valid and invalid groups to reduce the number of test cases?

A) Boundary Value Analysis

B) Equivalence Partitioning

C) Decision Table Testing

D) Error Guessing

✓ **Answer: B**

Explanation: Equivalence Partitioning groups inputs into classes that are expected to behave similarly, allowing representative test cases to be selected efficiently.

Analytical / Problem Solving MCQs (Q1–Q15)

Q1. If 5 machines produce 5 parts in 5 minutes, how many minutes will 100 machines take to produce 100 parts?

A) 5

B) 20

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C) 100

D) 500

✓ **Answer: A**

Explanation: Each machine produces one part in 5 minutes. Therefore, 100 machines produce 100 parts in the same 5 minutes.

Q2. Find the next number in the series:

2, 6, 12, 20, 30, ?

A) 40

B) 42

C) 44

D) 48

✓ **Answer: B**

Explanation: Differences are +4, +6, +8, +10, so the next difference is +12 $\rightarrow 30 + 12 = 42$.

Q3. Which number does not belong to the group?

A) 16

B) 25

C) 36

D) 45

✓ **Answer: D**

Explanation: 16, 25, and 36 are perfect squares; 45 is not.

Q4. If ALL BOOKS are PENS and ALL PENS are TABLES, then ALL BOOKS are:

A) Chairs

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- B) Tables
- C) Notebooks
- D) None

☒ **Answer: B**

Explanation: By transitive reasoning, Books $\subseteq$ Pens and Pens $\subseteq$ Tables, so Books $\subseteq$ Tables.

Q5. A train 120 m long crosses a pole in 6 seconds. What is its speed?

- A) 20 m/s
- B) 15 m/s
- C) 18 m/s
- D) 24 m/s

☒ **Answer: A**

Explanation: Speed = Distance $\div$ Time = $120 \div 6 = 20$ m/s.

Q6. Which data structure is most suitable for checking balanced parentheses?

- A) Queue
- B) Stack
- C) Linked List
- D) Tree

☒ **Answer: B**

Explanation: A stack follows LIFO, making it ideal for matching opening and closing brackets.

Q7. Find the missing number:

3, 7, 15, 31, 63, ?

- A) 95

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B) 127

C) 129

D) 131

✓ **Answer: B**

Explanation: Pattern: $(\text{Previous} \times 2) + 1$.

Q8. If the day after tomorrow is Friday, what day was yesterday?

A) Monday

B) Tuesday

C) Wednesday

D) Thursday

✓ **Answer: B**

Explanation: Day after tomorrow = Friday $\rightarrow$ Today = Wednesday $\rightarrow$ Yesterday = Tuesday.

Q9. A program has a time complexity of $O(n^2)$. If the input size doubles, the approximate number of operations becomes:

A) Double

B) Four times

C) Eight times

D) Sixteen times

✓ **Answer: B**

Explanation: $(2n)^2 = 4n^2$.

Q10. Which option completes the analogy?

CPU : Computer :: Engine : ?

A) Driver

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B) Vehicle

C) Fuel

D) Road

✓ **Answer: B**

Explanation: A CPU is the processing unit of a computer; similarly, an engine powers a vehicle.

Q11. If all roses are flowers and some flowers are red, which statement is definitely true?

A) All roses are red.

B) Some roses are red.

C) All roses are flowers.

D) No flower is red.

✓ **Answer: C**

Explanation: Only the first statement is guaranteed by the given information.

Q12. Which number should replace the question mark?

8 : 64 :: 12 : ?

A) 96

B) 120

C) 144

D) 156

✓ **Answer: C**

Explanation: $8^2 = 64$, so $12^2 = 144$.

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Q13. Which algorithmic approach is generally preferred when a problem contains overlapping subproblems and optimal substructure?

- A) Greedy
- B) Dynamic Programming
- C) Divide and Conquer
- D) Brute Force

 **Answer: B**

Explanation: Dynamic Programming stores solutions to overlapping subproblems to improve efficiency.

Q14. Three statements are given:

- All programmers are logical.
- Some logical people are gamers.

Which conclusion is definitely true?

- A) All programmers are gamers.
- B) Some programmers are gamers.
- C) All programmers are logical.
- D) No gamers are programmers.

 **Answer: C**

Explanation: The second statement does not establish any guaranteed overlap between programmers and gamers.

Q15. A binary search algorithm successfully finds an element in a sorted array of 1,024 elements. In the worst case, approximately how many comparisons are needed?

- A) 10
- B) 32
- C) 100

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D) 512

✓ **Answer: A**

Explanation: Binary Search has a worst-case time complexity of $O(\log_2 n)$. Since $\log_2(1024) = 10$, it takes at most about 10 comparisons.